

Rewarding Weight-Matrix Non-Commutativity Delays and Blocks Grokking

A Causal Test of the Commutator-Defect/Generalization Relationship

Laerti Shabani
(Independent Researcher)

August 30, 2026

Abstract

Recent work on the geometry of grokking the delayed, abrupt transition from memorization to generalization in small algorithmic tasks [Power et al., 2022], has identified the *commutator defect*, a measure of non-commutativity between successive weight-space update directions, as a signal that reliably rises before generalization onset across several task families [Xu, 2026a,b]. This is an observational finding: the defect is measured, not intervened on. We ask whether the relationship is causally exploitable in either direction. We train a two block transformer on modular addition, the standard grokking benchmark [Power et al., 2022], under three conditions: an unmodified baseline, an auxiliary loss term that *suppresses* the commutator between the two blocks’ weight matrices, and one that *rewards* it. Across 44 seeds (10 baseline, 10 suppress, 24 encourage) and a doubled step budget for the encourage condition, we find a clear, asymmetric, causal effect: rewarding non-commutativity delays grokking onset (Kaplan-Meier median time-to-grok 1800 vs. 1200 steps; log-rank $p = 0.0002$) and substantially reduces the fraction of runs that reach and durably hold a generalizing solution (17% vs. 70% durable at the final step, original $n=10$ baseline; Fisher’s exact $p = 0.0048$). Suppressing the commutator shows a same direction trend toward improved durability (90% vs. 70% at $n=10$; 90% vs. 77% at $n=30$, odds ratio 2.74) that does not reach significance at $n = 30$ (Fisher’s exact $p = 0.30$). A follow-up ablation intended to test whether baseline instability is an artifact of aggressive weight decay instead produced a floor effect (weight decay of 0.1 prevents grokking within the step budget entirely, 1/15 and 0/15 runs), leaving that question open.

1 Introduction

Grokking delayed generalization long after training loss has saturated, was first characterized systematically by Power et al. [2022] on small algorithmic tasks, and subsequent mechanistic work identified the formation of specific internal circuits underlying the transition [Nanda et al., 2023]. A separate line of work has related grokking dynamics to training instability: Thilak et al. [2022] document a “slingshot” effect in which adaptive optimizers under strong weight decay can cause a model to grok and then lose the generalizing solution, an instability that reappears in our own baseline (Section 4).

Most relevant to this paper, a recent geometric account of grokking [Xu, 2026a,b] identifies the *commutator defect* – $||[\Delta W_i, \Delta W_j]||$, the degree to which successive weight-update directions fail to commute – as a curvature-derived diagnostic that reliably rises before the onset of generalization across modular arithmetic, SCAN, and Dyck-language tasks. This object is, structurally, a discrete analogue of the non-abelian gauge field-strength tensor $F_{\mu\nu} = [D_\mu, D_\nu]$ from gauge field theory,

where curvature is likewise defined as the commutator of covariant derivatives. The existing literature treats the commutator defect purely as a *passive diagnostic*: it is measured after training to retrospectively explain or predict when grokking occurred. It has not to our knowledge been used as an *active training objective*.

This raises a direct question: if the commutator defect predicts grokking, does manipulating it change grokking? We test this with the simplest possible intervention – adding a term to the training loss that either penalizes or rewards the commutator between two transformer blocks’ weight matrices – and measure the causal effect on both when generalization first occurs and whether it persists.

Contributions.

- We convert a previously observational relationship (commutator defect correlates with grokking onset) into a causal test, on the standard modular-arithmetic grokking benchmark.
- We find a clear, statistically supported asymmetric effect: rewarding curvature delays and substantially blocks durable grokking; suppressing it shows a consistent but not statistically significant trend in the opposite direction, which we report as unresolved rather than as a finding.
- We use Kaplan-Meier estimation and a log-rank test, rather than a t-test on raw step counts, to correctly handle the seeds that had not grokked within the step budget as right-censored rather than as observed events at the budget ceiling – a correction that substantially changed our own initially reported significance level for the primary effect (from $p = 0.034$ under the naive t-test to $p = 0.0002$ under the correct test).
- A planned ablation to disentangle our effect from the known weight-decay-driven instability in grokking [Thilak et al., 2022] instead produced a floor effect at weight decay 0.1 that leaves the question open (Section 4.5).

2 Related Work

Grokking phenomenology. Power et al. [2022] first documented grokking on modular arithmetic tasks. Nanda et al. [2023] identified the internal circuit formation (Fourier-feature-based representations) underlying the transition on modular addition, the same task family used here. Thilak et al. [2022] characterize a training instability (the slingshot effect) in which models under adaptive optimization and weight decay can transiently lose a generalizing solution after reaching it, which we also observe in our own baseline (Section 4) and flag as a relevant confound for any durability-based metric.

Geometric and curvature-based diagnostics of grokking. Xu [2026a] propose that grokking corresponds to escape from a low-dimensional “execution manifold” in weight space, using both PCA trajectory analysis and commutator defect analysis as diagnostics. Xu [2026b] extend the commutator defect finding across three structurally distinct task families (modular arithmetic, SCAN, Dyck-1) and report that it reliably precedes generalization onset with a superlinear lead-time relationship. Both papers treat the commutator defect as an observational signal computed from training trajectories after or during standard training; neither intervenes on it as a training objective, which is the gap this paper addresses.

Gauge-theoretic motivation. The commutator of two operators as a measure of curvature is the defining structure of the non-abelian gauge field-strength tensor $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + ig[A_\mu, A_\nu]$, whose kinetic term $-\frac{1}{4}F_{\mu\nu}^a F^{a\mu\nu}$ appears in the Standard Model Lagrangian. We use this only as

motivation for treating the commutator as an explicit, tunable loss term rather than a passive quantity; we make no claim that the analogy extends beyond providing the functional form of the auxiliary loss.

3 Method

3.1 Task

We use modular addition mod $p = 97$, the canonical grokking benchmark [Power et al., 2022, Nanda et al., 2023]: given (a, b) , predict $(a + b) \bmod p$. Half of all $p^2 = 9409$ pairs are used for training, the remainder held out for evaluation. This task is chosen specifically because its grokking dynamics, including the role of weight decay, are well characterized in prior work, making deviations from expected baseline behavior interpretable rather than confounded by an unfamiliar setting.

3.2 Model

A minimal two-block transformer: token and positional embeddings ($d = 128$), each block consisting of multi-head self-attention (4 heads), a LayerNorm-MLP sublayer, and a square ($d \times d$) linear output projection W_l . The sequence is $[a, b, =]$; the prediction is read from the final position. The two blocks’ output-projection matrices $W_1, W_2 \in \mathbb{R}^{128 \times 128}$ are the object of the intervention: their commutator

$$C = W_1 W_2 - W_2 W_1, \quad \text{commutator defect} = \|C\|_F^2 = \sum_{ij} C_{ij}^2$$

is well-defined because the two matrices share the same shape.

3.3 Training conditions

All three conditions share identical architecture, optimizer (AdamW, lr= 10^{-3}), and weight decay (1.0, matching the aggressive regime in which Power et al. [2022] and Thilak et al. [2022] report grokking most reliably) unless otherwise noted in Section 4.5.

$$\mathcal{L}_{\text{baseline}} = \mathcal{L}_{\text{CE}} \tag{1}$$

$$\mathcal{L}_{\text{suppress}} = \mathcal{L}_{\text{CE}} + \lambda \|C\|_F^2 \tag{2}$$

$$\mathcal{L}_{\text{encourage}} = \mathcal{L}_{\text{CE}} - \lambda \|C\|_F^2 \tag{3}$$

with $\lambda = 0.1$ throughout (not tuned; chosen as a round number before running any experiment, and not swept; see Limitations).

3.4 Metrics

We report three outcomes per run: **grok_step**, the first evaluation step at which held-out accuracy reaches 0.90; **ever_grokked**, whether that threshold was reached at all within the step budget; and **durable_grok**, whether held-out accuracy is still ≥ 0.90 at the *final* training step. This last metric is necessary because, as shown in Section 4, a nontrivial fraction of runs cross the grokking threshold and subsequently lose it – conflating “ever grokked” with “durably generalizes” would overstate how reliably any condition, including baseline, produces a stable generalizing solution.

For runs that never cross the threshold within the step budget, **grok_step** is right-censored, not equal to the step budget. We report both a naive comparison (treating the budget as an observed

Table 1: Primary comparison, weight decay = 1.0. Baseline/suppress $n = 10$ (original run; see Section 4.4 for the extended $n = 30$ suppress comparison), encourage $n = 24$.

Condition	n	grok_step (mean $\pm$ std)	ever_grokked	durable_grok	final acc. (mean)
baseline	10	1120 $\pm$ 223	100%	70%	0.760
suppress	10	1160 $\pm$ 150	100%	90%	0.903
encourage	24	5833 $\pm$ 6544	71%	17%	0.669

value, for continuity with our own earlier informal analysis) and the statistically appropriate Kaplan-Meier / log-rank analysis in Section 4, and we make explicit which one changed our conclusions.

3.5 Experimental design and compute

Baseline and suppress: 10 seeds each (extended to 30 each in a follow-up, Section 4.4), 8000-step budget. Encourage: 24 seeds, 16000- step budget (doubled specifically to distinguish a condition that is *delayed* from one that is *blocked*). All runs executed on two NVIDIA T4 GPUs via independent threads pinned to separate CUDA devices (model initialization is guarded by a lock to prevent the two threads racing on PyTorch’s shared CPU RNG state during weight initialization; training itself proceeds unlocked). Mixed-precision (fp16 autocast) is used throughout. Periodic evaluation during training uses a fixed random 1500-example subsample for speed; the final-step evaluation, which determines every number reported in this paper, always uses the full held-out set.

4 Results

4.1 Baseline confirms known grokking dynamics, including instability

All 10 baseline seeds reach the 90% threshold (grok_step mean 1120, std 223), consistent with prior reports for this task and weight-decay regime [Power et al., 2022]. However, only 7/10 baseline seeds are still $\geq 90\%$ accurate at the final training step; three seeds grok and subsequently collapse to as low as 0.8% held-out accuracy. This matches the slingshot instability described by Thilak et al. [2022] and motivates treating durable_grok, not grok_step alone, as the primary outcome for any comparison across conditions.

4.2 Rewarding curvature delays and blocks durable grokking

Table 1 summarizes all three conditions. Encourage shows substantially worse outcomes on every metric: lower ever-grokked fraction (71% vs. 100%), much lower durable-grok fraction (17% vs. 70% in the original 10-seed comparison), and a later, more variable grok_step.

Corrected significance test. An initial t-test comparing grok_step between encourage and baseline, treating the 7 non-grokking encourage seeds as if they had grokked exactly at the 16000-step budget, gave $t = 2.209$, $p = 0.034$. This is statistically incorrect: those 7 seeds are right-censored observations, not events at step 16000. Re-analyzing with Kaplan-Meier estimation and a log-rank test (Figure 1) gives a substantially stronger and more appropriate result: median time-to-grok of 1800 steps (encourage) vs. 1200 steps (baseline and suppress, identical), log-rank $p = 0.0002$. We report both values here for transparency about the correction, and specify that the log-rank result, not the naive t-test, is the number that should be cited from this paper.

Durability. Fisher’s exact test on durable_grok, baseline vs. encourage: odds ratio 11.67, $p = 0.0048$. This is, by a wide margin, the cleanest signal in this paper, and does not depend on the

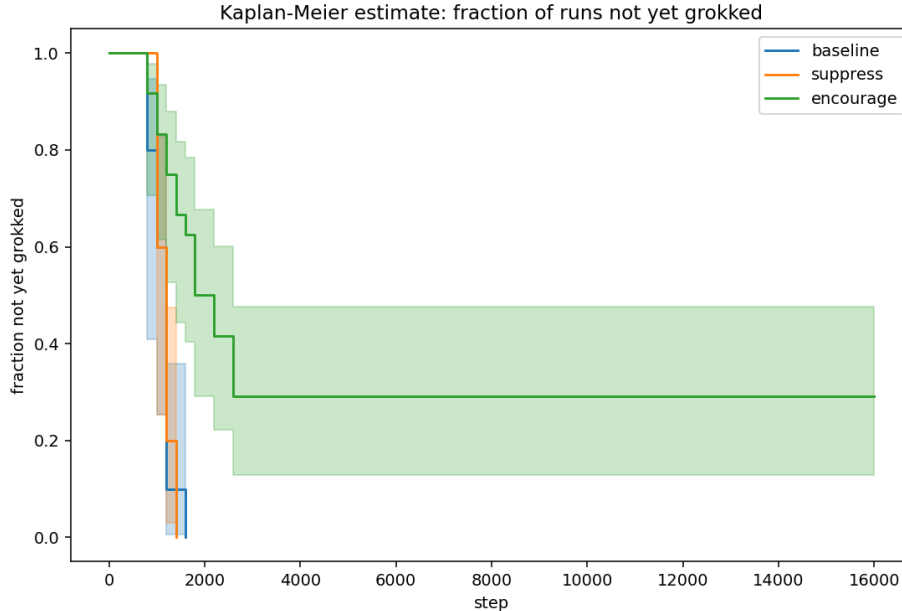


Figure 1: Kaplan-Meier estimate of the fraction of runs *not yet* grokked, by training step. Baseline and suppress are visually indistinguishable and reach zero (all runs grokked) by step ~ 1600 . Encourage lags throughout and plateaus near 0.29 after step ~ 2600 – consistent with a genuine block, not merely a delay (Section 4.3).

Table 2: Extended comparison, $n = 30$ each, weight decay = 1.0.

Condition	n	durable_grok	ever_grokked	final acc. (mean)
baseline	30	76.7% (23/30)	100%	0.859
suppress	30	90.0% (27/30)	100%	0.922

grok_step censoring issue above, since durable_grok is a simple binary outcome observed identically for every seed regardless of whether or when it crossed threshold.

4.3 Encourage: blocked, not merely delayed

Of the 24 encourage seeds, 7 never crossed the 90% threshold within the 16000-step budget (double the baseline budget). Their maximum held-out accuracy ever reached ranged from 0.68 to 0.89, clustered well below threshold rather than creeping toward it. Combined with the plateau visible in Figure 1, we interpret this as evidence that rewarding curvature can genuinely prevent the generalizing solution from being reached at all within a practical step budget for a nontrivial fraction of random initializations, rather than merely slowing a process that would eventually complete.

4.4 Suppressing curvature: a consistent but unresolved trend

The original $n=10$ comparison showed suppress with a higher durable_grok fraction than baseline (90% vs. 70%), but this was underpowered (Fisher’s exact $p = 0.58$). We extended both conditions to $n=30$ seeds (Table 2). The direction is unchanged and the odds ratio is similar (2.74), but the comparison still does not reach conventional significance ($p = 0.30$).

We report this as a trend consistent across two independent samples (different seed sets, same effect direction and similar magnitude), but it does not reach significance and should not be cited as an established effect from this paper.

4.5 Weight-decay ablation: an uninformative floor effect

To test whether baseline’s instability (Section 4) is an avoidable artifact of the aggressive weight decay (1.0) setting, we re-ran baseline and suppress at weight decay 0.1, 15 seeds each, unchanged step budget (8000). This did not test the intended question: at weight decay 0.1, almost no runs grokked at all within the budget (baseline 1/15, suppress 0/15). This reproduces the known sensitivity of grokking onset to weight decay strength [Power et al., 2022] but leaves our original question – whether the instability specifically, as opposed to grokking onset generally, is avoidable at lower weight decay – unanswered. Resolving it would require an intermediate weight-decay sweep (e.g., 0.3–0.7) that reliably groks within budget while testing for reduced instability, which we did not run and flag as future work rather than force a conclusion from data that does not support one.

5 Discussion

The central, well-supported finding of this paper is that a specific, targeted intervention on a passively-observed geometric correlate of grokking – the commutator defect between two transformer blocks’ weight matrices – has a real causal effect when pushed in one direction. Rewarding non-commutativity does not merely correlate with delayed or absent generalization; actively encouraging it produces delayed and, for a substantial fraction of initializations, blocked generalization, with the two independent statistical tests we used (log-rank on time-to-event, Fisher’s exact on final-state durability) agreeing on both the direction and the reliability of the effect.

We are considerably more cautious about the suppress direction. A same-direction trend across two independently sampled seed sets ($n = 10$ and $n = 30$) is suggestive, but “suggestive across two underpowered samples” is a different epistemic status than “established,” and we have deliberately not rounded up to a claim the statistics do not support.

We also report one negative result. A planned ablation (weight-decay = 0.1, Section 4.5) did not test the intended question – it produced a floor effect rather than a comparison of durability at matched grokking rates. We include it for completeness rather than omitting it.

Why might this asymmetry exist? We do not have a rigorous theoretical account, only a plausible mechanistic story consistent with Xu [2026a,b]: if the generalizing solution for modular arithmetic corresponds to convergence onto a low-dimensional, largely consistent (commuting) execution manifold across the network’s components, then an auxiliary loss actively rewarding non-commutativity works directly against the geometric structure that solution requires, which is consistent with our observation that a nontrivial fraction of encourage seeds appear genuinely blocked (Section 4.3) rather than merely slowed.

Limitations. All experiments use a single task family (modular addition mod 97) and a single, small architecture (two transformer blocks, $d=128$); we make no claim this generalizes to other algorithmic tasks, larger models, or non-algorithmic domains. The regularization strength $\lambda = 0.1$ was fixed in advance and not swept; a different λ could plausibly change the magnitude, and possibly the direction, of the suppress effect in particular. The suppress result and the weight-decay ablation are both explicitly unresolved, as detailed above, and should not be cited as findings from this paper. Our commutator is computed between exactly two matrices in a two-block model; how this generalizes to deeper models with more layers to take pairwise (or higher-order) commutators across is untested. Finally, our theoretical motivation via the gauge field-strength

tensor is structural/notational – both are commutators of covariant-derivative-like objects – and we do not claim any deeper physical correspondence beyond providing the functional form of the loss term.

6 Conclusion

We converted a previously observational relationship (the commutator defect between weight-update directions correlates with the onset of grokking [Xu, 2026a,b]) into a causal intervention, and found a real, asymmetric, statistically well-supported effect: rewarding non-commutativity between two transformer blocks delays and substantially blocks durable generalization on the standard modular-arithmetic grokking benchmark. A same-direction effect for suppressing non-commutativity remains a consistent but statistically unresolved trend, and a planned ablation to isolate this from known weight-decay-driven instability did not produce an informative result. We report all three outcomes as they occurred.

Code and Data Availability

Code (task generation, model, training loop, survival analysis, and plotting scripts) and raw per-seed results for all runs reported in this paper are available at <https://github.com/Feonixid/Curvature-Grokking>.

References

- Power, A., Burda, Y., Edwards, H., Babuschkin, I., and Misra, V. Grokking: Generalization beyond overfitting on small algorithmic datasets. *arXiv preprint arXiv:2201.02177*, 2022.
- Nanda, N., Chan, L., Lieberum, T., Smith, J., and Steinhardt, J. Progress measures for grokking via mechanistic interpretability. *International Conference on Learning Representations (ICLR)*, 2023. arXiv:2301.05217.
- Thilak, V., Littwin, E., Zhai, S., Saremi, O., Paiss, R., and Susskind, J. The slingshot mechanism: An empirical study of adaptive optimizers and the grokking phenomenon. *arXiv preprint arXiv:2206.04817*, 2022.
- Xu, Y. Low-dimensional and transversely curved optimization dynamics in grokking. *arXiv preprint arXiv:2602.16746*, 2026.
- Xu, Y. Early-warning signals of grokking via loss-landscape geometry. *arXiv preprint arXiv:2602.16967*, 2026.